

DiskPlay: Dynamic Projection Mapping on Rotating Platforms for Extended Holographic Display

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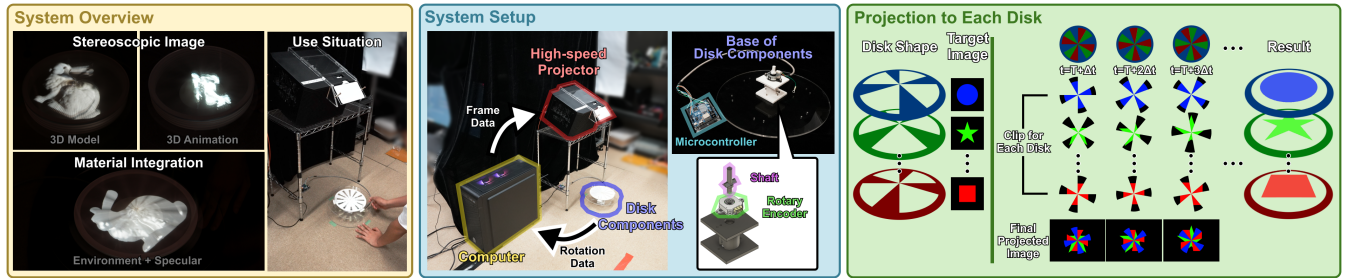


Figure 1: The overview of DiskPlay. Projection Result: Two types of projection results and use situation. **System Setup:** The system consists of a disk component, a computer, and a high-speed projector. The disk component transfers the rotary encoder values to the computer via a microcontroller. **Projection to Each Disk:** Clip the target image the user want to project on each disk along the disk angle and project the merged image.

Abstract

This paper presents DiskPlay, a projection-based platform designed for extended holographic displays through projection techniques on rotating multi-layered disks. This system performs dynamic projection mapping on rotating disks according to their shape, projecting a different image for each layer. It provides special visual effects such as stereoscopic images and interactions such as rearranging the disks or manually adjusting the rotation speed to change the image style. This paper details the implementation methodology, and showcases two types of applications, stacking with depth and interaction with rearranging disks. It also discusses the usefulness and limitations of the system.

CCS Concepts

• Human-centered computing → Displays and imagers.

Keywords

Dynamic Projection Mapping, Rotating Disk, High-speed Projector

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1 Introduction

Several previous studies have proposed projection systems that utilize rotating objects for expressing object deformation [Smoot et al. 2010], physically presenting materials [Yoshida et al. 2016], and projecting stereoscopic images [Katsuyama et al. 2023; WAKAT-SUKI et al. 2024]. These systems have the potential to create more expressive visual effects in the entertainment fields, and improve work efficiency through previsualization of the computer graphics. However, several challenges exist in introducing such systems to real-world applications. Firstly, the speed of rotation pose significant concerns. These systems typically require high rotation speeds (> around 600RPM), which raises safety issues. Secondly, some of them require complex rotational control equipment. Additionally, rotation speed of such systems tends to be predetermined by the projection controller, making it is challenging for users to adjust the rotation speed interactively. Thirdly, these systems often lack scalability. Altering the shape of the rotating object is generally not considered.

To address these limitations, we propose DiskPlay, a projection system that projects different images onto rotating multi-layer disks using a high-speed projector. This system can display stereoscopic images, and provide interactions such as changing the style of images by rearranging disks. The disk is designed with a specific cut pattern to enable projection at lower speeds of approximately 120 RPM. This system incorporates rotation speed detection to synchronize image projection, allowing the use of inexpensive motors with unstable rotational speeds or even hand-cranked mechanisms. This paper details the implementation of DiskPlay, followed by demonstrations of projecting a stereoscopic image and customizing multiple material elements through rearranging disks.

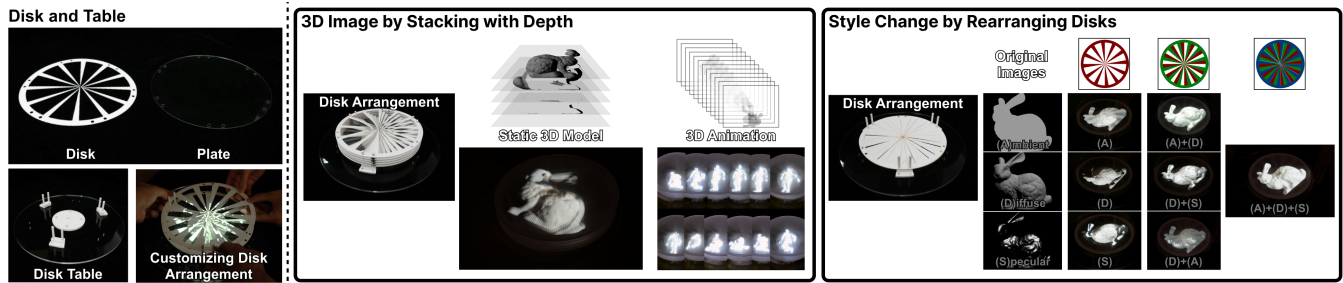


Figure 2: Disk table appearance and applications. *3D Image by Stacking with Depth*: Two examples of stereoscopic images, *Style Change by Rearranging Disks*: (A) Albedo, (D) Diffuse, and (S) Specular elements by Phong shading are visible and invisible depending on the number of disks.

2 Implementation

As shown in Figure 1 (center), DiskPlay primarily comprises disk components, a host computer, and a high-speed projector.

The disk components consist of a base part and a disk table. The shaft of the base part (Figure 1, center) is connected to an absolute rotary encoder (AMT22) underneath it. A microcontroller processes and decodes the signals from the encoder and sends the absolute angle of the table to the host computer. As shown in Figure 2 (left), the disks are fabricated by cutting paper. The number of disks is determined by the desired configuration, target rotation speed, and projector requirements. Based on preliminary testing, we selected a configuration of three or five disks, each shaped like a 12-blade fan, for manual rotation (approximately 120 RPM). The inner shape of each disk is offset by a regular angle to ensure non-overlapping configurations. These disks are stacked on a rotating table which is a larger plastic plate. They can also be stacked vertically by inserting spacers between them.

The host computer receives the angle data from the microcontroller via serial communication. As illustrated in Figure 1 (right), it renders a composite projected image by slicing the image that should appear on each disk according to its current angle. The image is rendered in real-time using openFrameworks¹. Dynafash projector² projects rendered images to the disks at 925 Hz color projection. As a result, different images are projected on each layer.

3 Applications

3.1 3D Image by Stacking with Depth

By mounting disks between spacers, DiskPlay makes it possible to project stereoscopic images. In this demonstration, a 3D model is segmented across multiple disks, allowing the viewer to perceive the stereoscopic image (Figure 2, center). By adjusting the number of disks, the depth resolution can be varied according to the use case. We tested configurations with five disks in this demonstration.

3.2 Style Change by Rearranging Disks

Another feature of DiskPlay is the ease of rearranging disks, allowing users to interactively customize the projected image. In this demonstration, the ambient, diffuse, and specular components are

decomposed and projected onto each disk (Figure 2, right). The user can select and integrate them by rearranging disks.

4 Discussion and Conclusion

Our demonstrations illustrate that stable projection can be maintained at speeds low enough for manual rotation and that a variety of images can be projected depending on the combination of disks. The disk shape used in this demonstration consisted of a fan shape with 12 blades based on preliminary testing. This configuration presents a trade-off: the lower the minimum rotation speed, the more blades in the disks are required, resulting in higher projection accuracy. Furthermore, the total delay from sensor detection to projection needs to be reduced further to prevent misalignment of the projection onto the disk.

This paper presents the DiskPlay platform, which allows users to experience extended projection with ease and accessibility. We aim to develop a system that can project images onto real-life rotating objects such as tires and DJ turntables. Future work includes implementing optical rotation detection using a camera and projecting with a commercial projector, which typically has lower refresh rates. Additionally, we intend to focus on developing a design system for visual effects and interactions possible on the DiskPlay platform.

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References

- Hidetaka Katsuyama, Shio Miyafuji, Arisa Kohtani, and Hideki Koike. 2023. Volu-Me Fan: Turn Your Layered-fan into a Volumetric Display. In *Proceedings of the 2023 ACM Symposium on Spatial User Interaction* (Sydney, NSW, Australia) (SUI '23). Association for Computing Machinery, New York, NY, USA, Article 39, 3 pages. <https://doi.org/10.1145/3607822.3618006>
- Lanny Smoot, Katie Bassett, Stephen Hart, Daniel Burman, and Anthony Romrell. 2010. An interactive zoetrope for the animation of solid figurines and holographic projections. In *ACM SIGGRAPH 2010 Emerging Technologies* (Los Angeles, California) (SIGGRAPH '10). Association for Computing Machinery, New York, NY, USA, Article 6, 1 pages. <https://doi.org/10.1145/1836821.1836827>
- Karin WAKATSUKI, Chiemi FUJIKAWA, and Makoto OMODANI. 2024. Universal Angle Visibility Realized by a Volumetric 3D Display Using a Rotating Mirror-Image Helix Screen. *IEICE Transactions on Electronics* E107.2, C (2024), 23–28. <https://doi.org/10.1587/transele.2023DII0003>
- Takatoshi Yoshida, Yoshihiro Watanabe, and Masatoshi Ishikawa. 2016. Phyxel: Realistic Display of Shape and Appearance using Physical Objects with High-speed Pixelated Lighting. In *Proceedings of the 29th Annual Symposium on User Interface Software and Technology* (Tokyo, Japan) (UIST '16). Association for Computing Machinery, New York, NY, USA, 453–460. <https://doi.org/10.1145/2984511.2984548>

¹<https://openframeworks.cc/>

²<https://us.teldevice.com/product/tb-6v-dynafash/>